

Review

Mammalian Polycistronic mRNAs and Disease

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Our understanding of gene expression has come far since the ‘one-gene one-polypeptide’ hypothesis proposed by Beadle and Tatum. In this review, we address the gradual recognition that a growing number of polycistronic genes, originally discovered in viruses, are being identified within the mammalian genome, and that these may provide new insights into disease mechanisms and treatment. We carried out a systematic literature review identifying 13 mammalian genes for which there is evidence for polycistronic expression via translation through an internal ribosome entry site (IRES). Although the canonical mechanism of translation initiation has been studied extensively, here we highlight a process of noncanonical translation, IRES-mediated translation, that is a growing source for understanding complex inheritance, the elucidation of disease mechanisms, and the discovery of novel therapeutic targets. Identification of additional polycistronic genes may provide new insights into disease therapy and allow for new discoveries of both translational and disease mechanisms.

One Messenger RNA, Multiple Polypeptides

George Beadle and Edward Tatum first proposed the model of gene expression in which individual genes encode single enzyme products, recast in subsequent years as the ‘one-gene one-polypeptide’ hypothesis [1,2]. Since then, in addition to the characterization of alternative splicing of transcripts to generate protein diversity, studies revealing several noncanonical translational mechanisms have shown that one gene may encode more than one functionally distinct polypeptide. One prominent example in viruses, protozoans, and invertebrates is the existence of numerous genes bearing multiple open reading frames (ORF; polycistronic) that encode two or more independently regulated proteins. Polycistronic genes in protozoans and invertebrates allow for translation initiation at two or more sites along a single mRNA transcript as an efficient means of coordinated gene expression. This expression strategy was not initially attributed to the vertebrate genome repertoire. Recently, a small but growing number of genes with either tandem or overlapping cistrons have been identified in vertebrates. Initiation of translation at a downstream or overlapping ORF is usually achieved by capitalizing on a highly structured stem-loop RNA element called an **IRES** (see [Glossary](#)) in the primary transcript, where ribosomes may bind downstream of the canonical translation start site. IRES function requires interaction with a distinct set of RNA-binding proteins (translation initiation factors and IRES *trans*-acting factors), which may allow differential, but coordinated, control of gene expression relative to the upstream ORF. Prominent mammalian polycistronic cellular genes appear to fall into at least four distinct categories based on mRNA structure and function of gene products: (i) two subunits of a multisubunit complex whose expression is coordinated in a single transcript; (ii) functionally similar gene products that are differentially co-expressed; (iii) functionally distinct gene products that have programmatically related expression; and (iv) signaling proteins generated by either stimulus-coupled protease cleavage or cap-independent translation ([Figure 1](#)). Here, we review

Trends

When compared with well-studied systems in viruses, protozoans, and invertebrates, recently identified mammalian polycistronic genes appear to maintain a similar level of coordinated expression control and offer a unique strategy for the temporal and spatial control of gene expression.

Studies of IRES translation in polycistronic mammalian genes demonstrate multiple possible functions of the resulting proteins.

Mutations that affect the expression of multiple cistrons and, therefore, different molecular pathways could explain some observations of genetic and clinical complexity.

Newly discovered polycistronic genes demonstrate that targeting of secondary proteins translated by IRES-mediated mechanisms could be a viable method of disease treatment using miRNA targeting or immunotherapy.

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our current understanding of mechanisms for bicistronic gene translation, examples of prominent polycistronic genes, and some clinical implications of genetic aberrations of these genes and therapeutic possibilities for treatment of the resulting disease symptoms.

Canonical Protein Synthesis in Eukaryotes

Translation of mRNA into protein involves three stages: initiation, elongation, and termination. The predominant form of translation initiation in eukaryotes, known as 'cap-dependent initiation', relies upon recognition of the m⁷GpppN(7-methylguanosine) 'cap' structure at the 5' end of the mRNA by a complex of canonical initiation factors (eukaryotic initiation factors; eIFs) termed 'eIF4F'. The eIF4F complex, which comprises eIF4A (a DEAD-box RNA helicase), eIF4E (the cap-binding protein), and eIF4G (a multidomain scaffold protein), recognizes and binds the 5'-cap. The eIF4F complex then recruits the 40S ribosomal subunit along with a second complex, the ternary complex (GTP-bound eIF2 and charged methionine initiator tRNA), as part of the 43S initiation complex, to form the 48S initiation complex. The 48S initiation complex then moves along the **5'-untranslated region (5'-UTR)** of the mRNA using the helicase activity of eIF4A to help unwind any potentially inhibitory secondary structures until an AUG in favorable conditions is encountered [3]. GTP is subsequently hydrolyzed to GDP in the presence of eIF5, followed by dissociation of some of the initiation factors. The 60S ribosomal subunit then joins the small subunit, resulting in an elongation-competent 80S ribosome. At this point, translation is initiated (Figure 2). The elongation phase of translation then proceeds until the ribosome encounters a termination codon [4,5].

Cap-Independent Translation in Vertebrates

Based on their primary nucleotide sequences, many mRNAs undergo intramolecular base pairing to produce highly ordered secondary structures capable of forming complex stem-loop conformations that facilitate the binding of specific proteins with functional consequences. For viral and eukaryotic mRNAs, highly structured IRES elements can form complexes with ribosomes together with a subset of the canonical eIFs and other RNA-binding proteins called **IRES trans-acting factors (ITAFs)** to allow for protein synthesis to initiate under conditions not favorable for (or not requiring) cap-dependent initiation (Figure 2) [6–8]. The eIF and ITAF requirements differ among IRESs. Interactions of eIFs and ITAFs with ribosomal subunits are thought to allow for the correct positioning of the IRES-mediation translation initiation codon at the ribosomal P-site [6]. However, extensive studies on initiation complexes for the IRES region have not been conducted. Once thought to be a property exclusive to viruses, approximately 10–15% of vertebrate cellular mRNAs are now believed to be capable of undergoing IRES-mediated protein synthesis based on the presence of IRESs located within the 5'-UTR downstream of the cap site [9]. One hundred and fifteen genes containing cellular 5'-UTR IRESs have been annotated, and a recent systematic study provides supporting evidence for known IRESs and suggests potentially novel IRESs using oligonucleotides verified for cap-independent translation activity [10,11].

Based on the functions of the genes studied to date, it appears that the 5'-IRES provides a means of expressing certain key proteins under conditions of cell stress when cap-dependent translation is inhibited. Alternative translation by cellular 5'-UTR IRESs may also occur during phases of angiogenesis, mitosis, and cell proliferation, particularly in genes that have IRESs located in the 5'-UTR region, such as *myc* family genes, and those encoding cyclin-dependent kinase 11 and vascular endothelial growth factor (*VEGF*) [6]. Structures of eIFs, and ITAFs have been defined for only a small number of these cellular IRESs. However, in several cases, IRES activities appear to be dependent on ITAF expression levels and localization, suggesting a strict regulation of polycistronic genes [6]. Unlike their viral counterparts, which comprise distinct families of structurally related IRESs, cellular IRESs appear to be defined by short motifs and ITAFs [8]. Thus, cellular IRESs have differentiated structures that potentially allow for the diverse functions and localization of the protein products produced under their control.

Glossary

(See Table 1, main text for additional explanations of gene and protein functions.)

5'-untranslated region (5'-UTR): region within the mRNA upstream of the start codon of coding sequence, often with complex secondary structures that may contain regulatory elements for canonical translation or an IRES.

CACNA1C calcium channel-associated transcription regulator (CCAT): transcription factor encoded by the *CACNA1C* gene.

Calcium voltage-gated channel subunit alpha1 A (CACNA1A): encodes the alpha1 A subunit of a calcium channel and the alpha1 ACT transcription factor.

eIF4E-binding protein (4E-BP): binds eIF4E preventing interaction between eIF4E and eIF4G and thereby preventing cap-dependent translation initiation

Free fatty acid receptor 1 (FFAR1): encodes the G-protein coupled receptors GPR40 and GPR41

Internal ribosome entry site (IRES): a specialized region within mRNA that can initiate translation independent of a 5' cap

IRES trans-acting factor (ITAF): a biomolecule involved in the regulation of the IRES

LAMB1: encodes Laminin B1, a membrane glycoprotein associated with cancer invasion

Melanoma-overexpressed antigen (meloe): encodes three antigens expressed in melanocytes and overexpressed in melanomas

Myotrophin (MTPM): in addition to encoding myotrophin, which allows dimerization of NF- κ B in cardiac tissue, the gene also encodes the antigen MPD6.

Poly(A)-binding protein (PABP): involved in the circularization of mRNA through its interaction with the poly(A) tail and eIF4G.

Repeat associated non-ATG (RAN) translation: specialized, novel method of translation that appears to require codon repeats for initiation.

Spinocerebellar ataxia type 6 (SCA6): ataxia caused by a polyglutamine repeat expansion in the voltage-gated Ca²⁺ channel, *CACNA1A*.

1-[1-(2-Thienyl)Cyclohexyl] piperidine (tenocyclidine)-binding protein gene (TCP-BP): a protein

Polycistronic Cellular Genes

Nearly all positive-sense RNA viruses have genomic RNAs that encode multiple protein products in the form of precursor polyproteins that are then processed to the functional polypeptides used by the virus during infection. A subset of these viruses (e.g., human rhinovirus, hepatitis C virus, and cricket paralysis virus) contains at least one ORF preceded by IRES structures. However, there is growing evidence that some cellular and vertebrate mRNAs also have IRES-like structures further downstream of the 5'-UTR, within or after the 5' proximal ORF, enabling the expression of proteins from tandem or overlapping ORFs. As with prokaryotic polycistronic genes, canonical translation typically initiates near the 5' ends of mRNAs of vertebrate bicistronic genes. However, in a few cases, it has been documented that the synthesis of a second protein is initiated via an IRES element located downstream of or within the first ORF.

comprising two subunits that forms a complex of proteins that bind glutamate agonists.

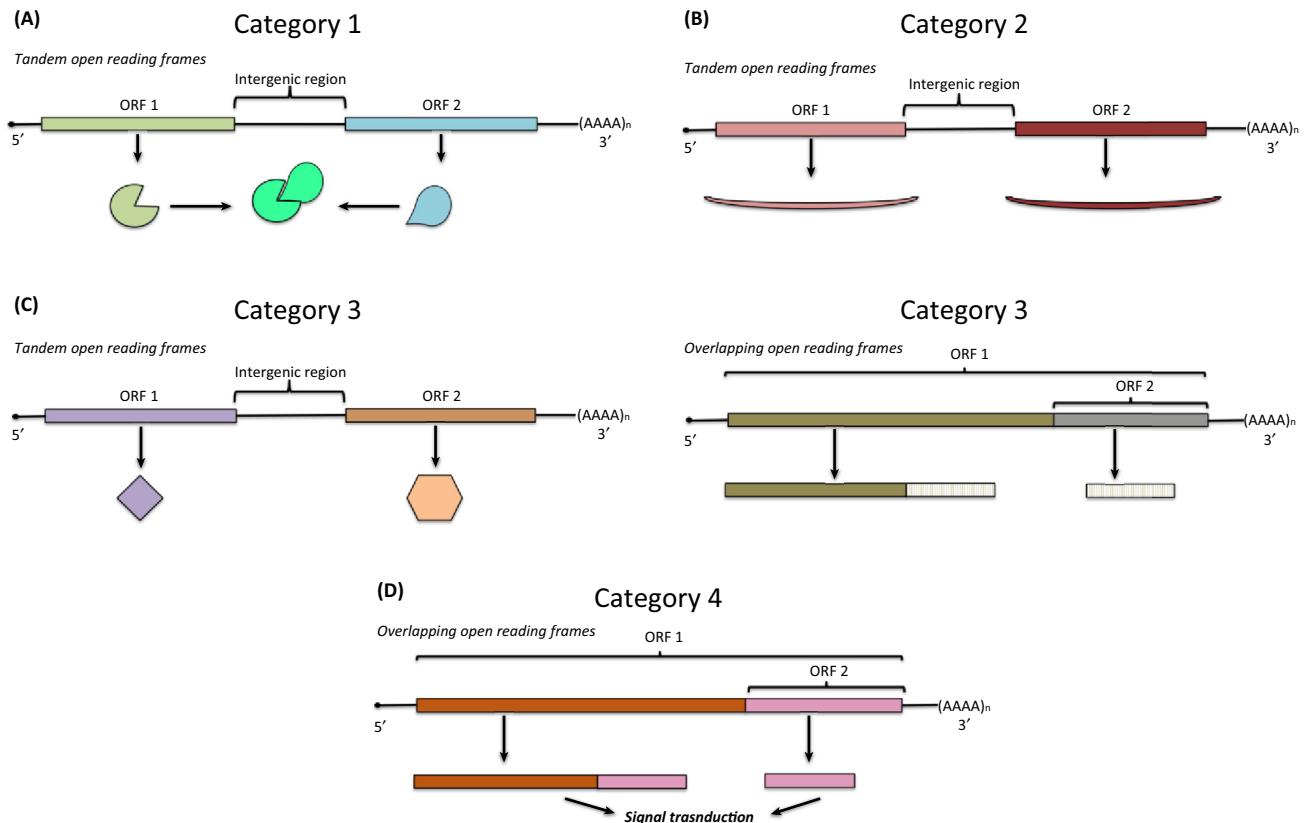
Due to multiple mechanisms of alternative gene expression and translation in eukaryotic cells, the identification of mRNAs harboring bona fide IRES sequences requires several stringent criteria that must be satisfied. Cryptic promoters in reporter plasmids and alternative splicing events that can lead to distinct transcripts and secondary protein products need to be ruled out [12,13]. In addition, multiple mechanisms of secondary protein translation, including ribosomal scanning, re-initiation, stop codon read-through, or translational frameshifting, may be found in the same gene. Conclusive evidence for the presence of an IRES requires exclusion of these mechanisms and several functional studies (Figure 3).

Regulation of Cellular IRES Activity

Although the mechanism of cellular IRES function is not well understood, there is increasing interest in identifying cell-specific expression levels of ITAFs and structural elements that define IRES activity. ITAFs have not been implicated in cap-dependent mechanisms and have been suggested to be specific for cap-independent translation [14]. As a result, ITAFs have an important role in determining the cell- or tissue-specific expression of polycistronic transcripts with IRESs. For example, in the gene encoding **melanoma-overexpressed antigen** (*meloe*), two IRESs encode T cell antigens that are selectively expressed in melanoma cells but are absent in melanocytes, suggesting that ITAFs are silenced in normal melanocytes [15,16]. In tissues that do not express or sequester ITAFs critical for IRES function, it would be expected that the polycistronic gene would function in a monocistronic manner, highly reliant on cap-dependent translation. Therefore, polycistronic activity can alter gene expression diversity merely through the availability of these IRES-specific factors [6]. However, loss-of and gain-of IRES-mediated translation may contribute to a pathophysiology in cases where disease tissue changes the availability of ITAFs to abnormally promote or eliminate IRES activity. **LAMB1**, for example, encodes the membrane glycoprotein Laminin B1, which has a significant role in cancer invasion via a 5'-UTR IRES regulated by the ITAF La. La expression, and subsequently IRES-mediated translation of Laminin B1, is decreased in normal hepatocytes compared with hepatocytes after the epithelial-mesenchymal transition (EMT), a critical step in the acquisition of invasive properties by carcinoma cells [17].

In addition to expression levels, the subcellular localization of the ITAFs is important for cellular IRES function [6,14,18]. A nuclear ITAF may sequester the mRNA until signaling changes allow for translocation to the cytoplasm for translation. Unbound ITAFs may also be compartmentalized until certain cues lead to translocation events necessary for translation of the IRES (e.g., the movement of an inhibitory ITAF from the cytoplasm to the nucleus) [6,14]. These hypotheses suggest that a separation of ITAFs from mRNA via subcellular compartmentalization is a critical regulatory mechanism for polycistronic mRNAs.

Lastly, the structural elements of the IRES are important for understanding their function and role in disease. miRNA, for example, have been shown to bind to a site in the 3'-UTR and act as



Trends in Genetics

Figure 1. Functional Organization of Bicistronic Genes. (A) Two subunits of a multisubunit complex whose expression is coordinated in a single transcript: each open reading frame (ORF) encodes a specific subunit of a larger protein complex. (B) Functionally similar gene products that are differentially co-expressed: a primary protein is expressed through canonical cap-dependent translation, while a secondary and functionally similar protein is differentially expressed through a cap-independent mechanism. (C) Functionally distinct gene products that have programmatically related expression: expression of two differentially functioning proteins is coupled with their operation in a particular biological pathway. Category 3 appears with both tandem and overlapping reading frames. (D) Signaling proteins generated by stimulus-coupled protease cleavage or by cap-independent translation: two overlapping ORFs encode necessary products for signal transduction, the primary product is a receptor that initiates signal transduction upon ligand binding, while the secondary product is a constitutively active signal.

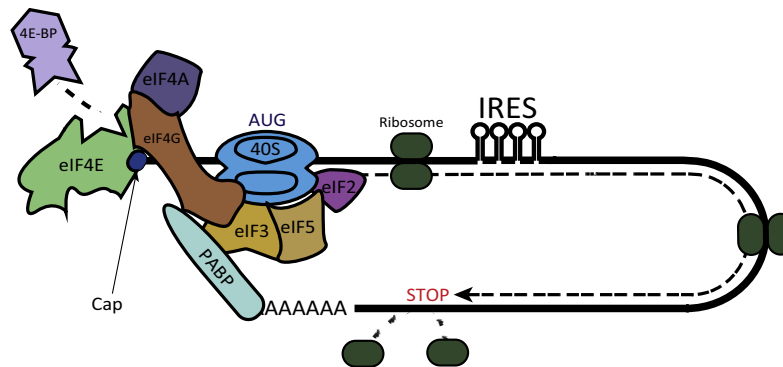
regulators of IRES-mediated translation for the *VEGFA* gene, which contains two IRESs in its 5'-UTR [19]. Moreover, cellular IRESs have been categorized into those that depend on short motifs and those that require an overall complex secondary or tertiary structure [11]. This categorization suggests that mutations that disrupt either short motifs or RNA structure would lead to a disruption in the normal function of the IRES and could lead to disease.

Why Polycistronic Genes?

Cellular polycistronic genes have several potential advantages for coordinated gene expression, organized here into four distinct categories that indicate expression mechanisms of each polycistronic gene (Figure 1). Furthermore, while polycistronic gene organization allows a unique and selective mechanism for the control of protein expression, this genetic strategy has several potential adverse clinical outcomes in the presence of genetic mutations or dysregulation of the IRES. Distinct mutations of polycistronic genes lead to complex and different phenotypes, possibly because of their effects on either the ORF or the IRES sequence itself. As a result, polycistronic genes could also open the door to novel therapies.

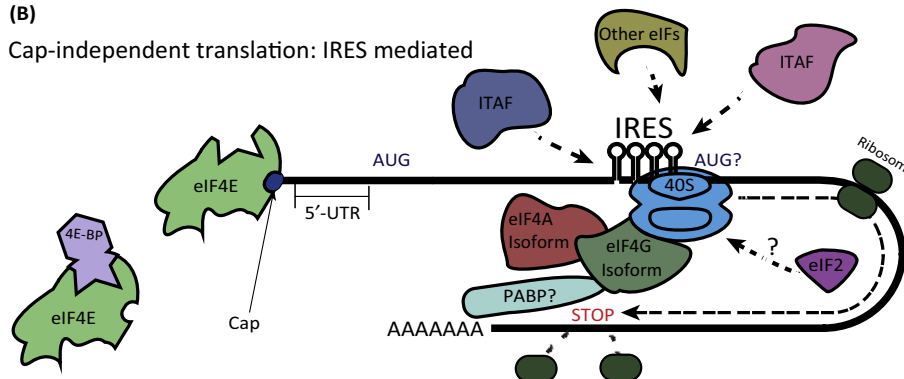
(A)

Cap-dependent translation



(B)

Cap-independent translation: IRES mediated



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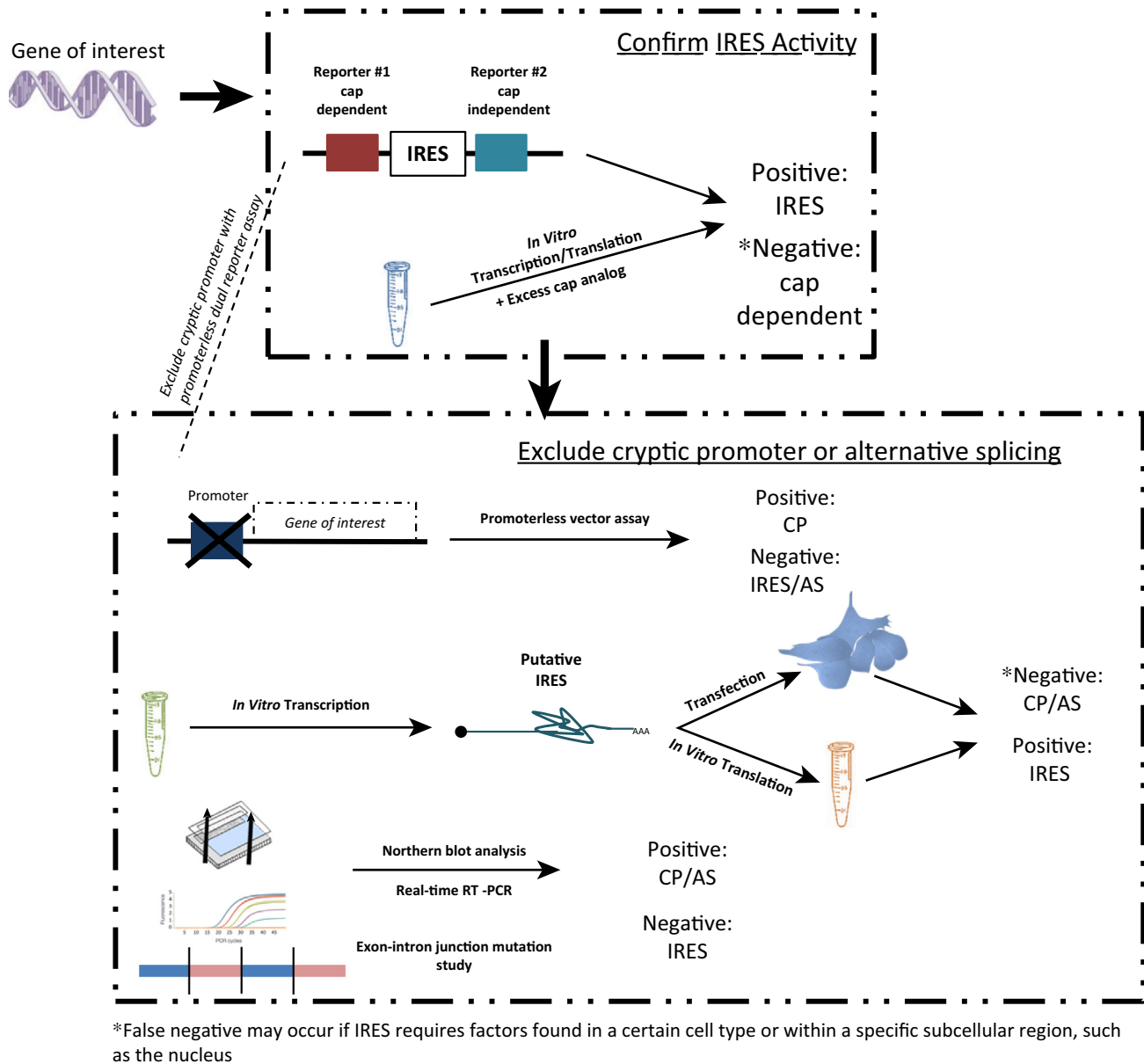
Figure 2. A Brief Overview of (A) Cap-Dependent Translation and (B) Internal Ribosome Entry Site (IRES)-Mediated, Cap-Independent Translation. Mammalian cap-dependent translation functions through the use of all the canonical initiation factors as well as circularization of the mRNA through the interaction of **poly(A)-binding protein (PABP)** and eukaryotic initiation factor 4G (eIF4G). Key initiation factors are shown here. Cap-independent IRES-mediated translation is thought to require a subset of IRES *trans*-acting factors (ITAFs) and an unknown number canonical initiation factors, and may also require circularization. Abbreviation: 4E-BP, eIF4E-binding protein.

Table 1 lists 13 polycistronic genes identified through the literature. Genes were not included that merely expressed truncated forms of the same protein with essentially the same function. Genes for which a cap-independent expression mechanism had not been supported experimentally were also excluded. In the following sections, we provide examples of the four functional classes of polycistronic gene and the biology and/or gene expression patterns associated with them (additional genes are mentioned in Table 1, Key Table).

Two Subunits of a Multisubunit Complex Whose Expression Is Coordinated in a Single Transcript

TCP-BP

1-[1-(2-Thienyl)Cyclohexyl]piperidine (tenocyclidine or TCP)-binding protein (TCP-BP) was identified as one of a complex of four proteins purified from rat brain synaptic membranes that bind glutamate agonists. TCP-BP comprises two subunits, PRO-1 and PRO-2, which are co-expressed subunits encoded by two tandem ORFs within a single bicistronic transcript. An IRES element preceding the second ORF that leads to translation of the PRO-2 subunit has been identified [20]. PRO-1 translation begins at an AUG at the start of ORF1, present 160 base pairs downstream of the mRNA start site. PRO-2 translation occurs from ORF2, 400 base pairs



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Figure 3. Key Methodology for Identifying Internal Ribosome Entry Site (IRES)-Mediated Translation. Confirmation of IRES activity is done by either: (i) insertion of the putative IRES sequence into a dual reporter vector; or (ii) *in vitro* transcription and translation with excess cap analog. Typical steps for exclusion of cryptic promoter (CP) activity or alternative splicing (AS): (i) promoterless vector assay; (ii) *in vitro* transcription of a single mRNA, and subsequent *in vitro* translation or cellular transfection; (iii) northern blotting and reverse transcription PCR spanning the exons.

downstream of ORF1, in the intergenic region of the mRNA. Both PRO-1 and PRO-2 are expressed in neurons in brain and show a tendency to form a complex required for TCP binding. Thus, PRO-1 and PRO-2 are formed through coordinately controlled, independent translation of two ORF regions via both a cap-dependent mechanism (PRO-1) and an IRES element (PRO-2). This rat brain gene reveals a potentially rare class of mammalian bicistronic genes in which two subunits of a multiprotein complex are translated by different expression strategies. There have yet to be any clinical associations with this bicistronic gene.

Key Table

Table 1. Mammalian Polycistronic Genes^a

Gene	Protein(s)	Functional category	Primary protein function	Secondary protein function	Disease relevance	Mechanism of expression of second cistron	Bicistronic organization	Refs
<i>TCP-BP</i>	PRO1 (8.9 kDa); PRO2 (9.5 kDa)	1	PRO1: interacts with PRO2 to form a low-affinity phencyclidine-binding entity that is part of a synaptic membrane complex	PRO2: interacts with PRO1 to form a low-affinity phencyclidine-binding entity that is part of a synaptic membrane complex	None reported	Putative IRES sequence between PRO1 ORF and PRO2 ORF	Tandem	[20]
<i>FFAR1</i>	GPR40 (31 kDa); GPR41 (38 kDa)	2	GPR40: long-chain fatty acid receptor	GPR41: short-chain fatty acid receptor	None reported	IRES in coding region of FFAR1 encodes GPR41	Tandem	[21–24]
<i>meloe</i>	MELOE-1 (4.3 kDa); MELOE-2 (5.1 kDa); MELOE-3 (5.9 kDa)	2	MELOE-3: no known function; poor immunogenicity, but is expressed in melanoma cells and normal melanocytes	MELOE-1: no known function; T cell antigen selectively expressed in melanoma cells MELOE-2: no known function; T cell antigen selectively expressed in melanoma cells	Potential target for T cell immune therapy for melanoma	IRES upstream of ORFs of MELOE-1 and MELOE-2; MELOE-3 produced by cap-dependent translation	Tandem	[15,16,25]
<i>MTPN</i>	Myotrophin (12.9 kDa); MPD6 (6.4 kDa)	3	Myotrophin: allows dimerization of NF- κ B in cardiac tissue	MPD6: no known function; antigen upregulated in response to interferon- α (IFN- α)	Myotrophin induces cardiac hypertrophy; MPD6 elicits humoral immune response in some cancers	MPD6 antigen ORF is located in the 3'-UTR region of the MTPN gene and is translated by an IRES located upstream	Tandem	[30]
<i>Cx43</i>	Cx43 (43 kDa); Cx43-CT (20 kDa)	3	Cx43: highly abundant connexin with junctional	Cx43-CT: implicated in autoregulating trafficking of Cx43	Hypoplastic left heart syndrome 1, oculodentodigital	Putative IRES or Cap-dependent internal translation	Overlapping	[45,46]

Table 1. (continued)

Gene	Protein(s)	Functional category	Primary protein function	Secondary protein function	Disease relevance	Mechanism of expression of second cistron	Bicistronic organization	Refs
			and nonjunctional functions		dysplasia, syndactyly type III; craniometaphyseal dysplasia			
<i>Cx55.5</i>	Cx55.5 (56 kDa); CT-11 (16 kDa)	3	Cx55.5: a connexin exclusively expressed in horizontal cells	CT-11: has nuclear translocation and may be involved in regulating light-dependent plasticity	Mutations in human homolog associated with cataracts	Putative IRES upstream of ATG at nucleotide position 1201	Overlapping	[47]
<i>pVHL-19</i>	pVHL-30 (30 kDa); pVHL-19 (19 kDa)	3	pVHL-30: tumor suppressor residing in cytosol	pVHL-19: comparable tumor suppressor in nucleus and cytoplasm with possibly different molecular substrates	von Hippel-Lindau disease	Mechanism not investigated	Overlapping	[48,49]
<i>CdcL1/CdcL2</i> (PITSLRE genes)	p110 (PITSLRE) (110 kDa); p58 (PITSLRE) (58 kDa) cyclin-dependent kinase isoforms	3	p110 (PITSLRE): regulates transcriptional processing by binding elongation factors and RNA-binding proteins	p58 (PITSLRE): binds cyclin D3 in G2/M phase of cell cycle	None reported	Increased translation of p58 during G2/M phase due to Unr and phosphorylated eIF-2 α binding via IRES near C-terminal end of p110	Overlapping	[26]
<i>SNRPN/SNURF</i>	SmN (24.6 kDa); SNURF (8.4 kDa)	3	SmN: splicing factor important for RNA processing	SNURF: implicated in the regulatory mechanism behind gene imprinting	Prader-Willi syndrome; Angelman syndrome	Mechanism not investigated	Tandem	[50]
<i>GDF1-LASS1</i>	GDF1 (39 kDa); LASS1 (39 kDa)	3	GDF1: TGF β superfamily member implicated in cell growth and differentiation with role in left-right	LASS1 (CerS1) ceramide synthase: TGF β superfamily member implicated in cell growth and differentiation; catalyzes synthesis of	GDF1: cardiovascular malformations; LASS1: progressive myoclonic epilepsy-8 (EPM8)	Mechanism not Investigated	Tandem	[51]

Table 1. (continued)

Gene	Protein(s)	Functional category	Primary protein function	Secondary protein function	Disease relevance	Mechanism of expression of second cistron	Bicistronic organization	Refs
			patterning and mesoderm induction	C18 ceramide in a fumonis B1-independent manner				
<i>CACNA1A</i>	α 1A (220 kDa); α 1ACT (75 kDa)	3	α 1A: voltage-gated calcium channel subunit involved in pre- and postsynaptic Ca^{2+} signaling, gene expression	α 1ACT: transcription factor important for expression of neural and Purkinje cell development genes	SCA6, episodic ataxia type 2 (EA-2), familial hemiplegic migraine type 1 (FHM-1), epilepsy	α 1ACT is made via an IRES in the C-terminal coding region of <i>CACNA1A</i>	Overlapping	[27–29]
<i>Notch2</i>	Notch2 (265 kDa); N2ICD (v-Notch2) (70 kDa)	4	Notch2: cell surface transmembrane receptor involved in cell fate decision during development by release of N2ICD	N2ICD: intracellular domain is decoupled and independent of Notch2 surface ligand	Hajdu-Cheney syndrome; Alagille syndrome	Proposed dual mechanism of second cistron initiation: ribosomal reinitiation and IRES both contribute equally to expression of N2ICD	Overlapping	[31–37]
<i>Her2</i>	HER2 (185 kDa); HER2 CTF (70 kDa)	4	HER2: tyrosine kinase receptor canonically known to act in secretory and endocytic pathways; also acts within the cell nucleoplasm and nucleus	HER2 CTFs: C-terminal fragments that encompass the transmembrane and cytoplasmic domains of full-length HER2 are associated with nodal metastasis, and localize to the cytoplasm and nucleus of cells	Hereditary breast cancer, cancer metastasis, tumor development	Hypothesized IRES within coding region (not confirmed); two methionines at positions 611 and 687 act as initiators of translation for CTFS	Overlapping	[38]

^aAbbreviations: *Cx43*, connexin 43; *Cx55.5*, connexin 55.5; *GDF1*, growth differentiation factor 1; *Her2*, human epidermal growth factor receptor 2; *LASS1*, ceramide synthase 1, also longevity assurance gene 1 protein homolog 1; *NOTCH2*, neurogenic locus notch homolog protein; *pVHL*, Von Hippel-Lindau tumor suppressor; *SNRPN*, small nuclear ribonucleoprotein polypeptide N; *SNURF*, SNRPN upstream reading frame.

Functionally Similar Gene Products that Are Differentially Co-Expressed

Two Proteins with Similar Structure and Function Are Differentially Expressed in the Same Biological System

FFAR1: GPR40 and GPR41

GPR40 and GPR41 are encoded by the bicistronic transcript **Free fatty acid receptor 1** (FFAR1), with GPR40 expression driven by a cap-dependent mechanism, whereas GPR41 expression is regulated via an IRES [21]. GPR40, a medium and long-chain fatty acid receptor, causes increased G α q/11-mediated intracellular calcium levels that stimulate the release of insulin in the presence of long-chain fatty acids [22]. GPR40 is expressed in a highly selective pattern, primarily in pancreatic beta and gastrointestinal tract cells [23]. GPR41 is also coupled to G α q/11, but is sensitive to short-chain fatty acids. Although its role in the metabolic pathways of beta cells has not yet been fully characterized, GPR41 has been implicated in energy homeostasis. Both proteins are also co-expressed in adipose tissue and the peripheral nervous system [23,24]. In beta cells, the expression of both receptors may be necessary to affect metabolic pathways. Dysregulation of either or both of the cistrons may affect the pathology of metabolic disease. It has been hypothesized that co-expression of these fatty acid-activated receptors could sensitize beta cells to different types and levels of fatty acid [21].

meloe: MELOE-1, MELOE-2, and MELOE-3

Cells of the melanocyte lineage express two polypeptides of unknown function, MELOE-1 and MELOE-2, from tandem cistrons in the *meloe* mRNA. Specific T cell clones recognize melanoma cells but not normal melanocytes based on the expression of one of these melanoma antigens on the surface of the melanoma cells [15,25]. Therefore, MELOE-1 and MELOE-2 have been studied for potential use in T cell immunotherapy in metastatic melanoma [15]. IRES sequences in the *meloe* mRNA were found upstream of both ORFs and it was speculated that the different translational mechanisms of 5' cap-dependent and IRES-mediated translation might be responsible for the selective expression of MELOE-1 and MELOE-2 only in melanoma cells [15]. Differential expression of ITAFs between normal melanocytes and melanoma cells could also account for the activation of these antigens selectively in cancer cells [15]. Furthermore, an additional ORF was identified in the 5' region of *meloe* encoding MELOE-3, translated by a cap-dependent pathway [16]. MELOE-3 was found to have poor immunogenicity, suggesting the importance of targeting the IRES-dependent proteins MELOE-1 and MELOE-2 for the immunotherapy of melanoma.

Functionally Distinct Gene Products that Have Programmatically Related Expression

Two Proteins with Distinct Functions Are Coupled for Co-Expression in the Same Biological Pathway

CdcL1/CdcL2 (PITSLRE genes): p110 and p58

PITSLRE/CDK11 duplicate genes *CdcL1* and *CdcL2* each encode two cyclin-dependent protein kinase isoforms, p110 and p58. The full-length *CdcL1/CdcL2* transcripts encode p110 proteins. p58 is produced from an ORF at the C-terminal end of the main ORF via IRES-mediated translation [26]. p110 is a known transcriptional regulator that regulates RNA-binding proteins and elongation factors. Furthermore, p110 was detected in all phases of the cell cycle [26]. Meanwhile, p58 binds cyclin D3 in the G2/M phase of the cell cycle [26]. Since translation of p58 is upregulated during the G2/M phase, the transcriptional activity of p110 during the cell cycle is likely coupled with the selective expression of p58 in the G2/M phase. Thus, these two PITSLRE genes encode functionally distinct proteins that are programmatically related and may undergo coupled expression. Although no specific disease association has been found with *CdcL1* and *CdcL2*, cyclin-dependent protein kinases carry out critical phosphorylation steps during the cell cycle that are crucial for proper cell function.

CACNA1A: α 1A and α 1ACT

At least one member of the family of genes encoding voltage-gated Ca^{2+} channels, **calcium voltage-gated channel subunit alpha1 A** (*CACNA1A*), is bicistronic. *CACNA1A* encodes the pore-forming calcium channel subunit, α 1A, through canonical, cap-dependent translation and a novel transcription factor, α 1ACT, through IRES-mediated translation from at least one spliced form of the same *CACNA1A* mRNA [27,28]. The α 1A voltage-gated Ca^{2+} channel is highly expressed in axon terminals of neurons, particularly in the cerebellum, where it mediates calcium entry to facilitate vesicular neurotransmitter release. α 1ACT translocates to the nucleus and binds to enhancer sequences containing an AT-rich motif to increase the expression of several genes expressed in Purkinje cells. α 1ACT expression enhances Purkinje cell dendritic growth and cerebellar cortical development. The α 1A mRNA bears a CAG repeat encoding a polyglutamine (polyQ) tract that is expanded in the autosomal dominant disease, **spinocerebellar ataxia type 6** (SCA6). SCA6 results from selective degeneration of Purkinje cells. α 1ACT bearing an expanded polyQ tract has reduced transcription factor function and is toxic to neuronal cells [27]. Selective suppression of α 1ACT by targeting the IRES (e.g., via RNA interference with miRNA) in adults before disease onset may be a therapeutic strategy for SCA6 [29].

MTPN: Myotrophin and MPD6

The gene, *MTPN*, encodes two proteins from a tandem bicistronic transcript. **Myotrophin** is translated by canonical cap-dependent translation, and MPD6 is produced from a second cistron via IRES-mediated translation. As with *CACNA1A*, the functions of the two proteins encoded by *MTPN* appear to be distinct, but programmatically related. The primary protein product, myotrophin, functions in the dimerization of NF- κ B in cardiac tissue. MPD6 is an antigen upregulated in response to interferon- α (IFN- α) [30]. As a result, MPD6 is implicated in the immune response in a subset of patients with polycythemia vera, chronic myelogenous leukemia, or prostate cancer [30]. Due to the possibility of IFN- α selectively increasing IRES translation and the implications for immunotherapy, this presents a striking similarity to the selective regulation of *meloe* IRESs in cancer cells. Different ITAF expression between melanocytes and melanoma cells has been attributed to different cistron expression in *meloe*. In the case of MPD6, this differentiation appears to be attributable to a response to the production of IFN- α by host cells. Thus, selective regulation of expression in the bicistronic gene presents a more complex view of the role of *MTPN* in normal and abnormal cellular function.

Signaling Proteins Generated by Stimulus-Coupled Protease Cleavage or by Cap-Independent Translation

The Secondary Protein Normally Coupled to Expression and Ligand Binding of the First Protein Has a Constitutively Active Route for Expression in Physiological or Pathological States

NOTCH2: Notch2 and N2ICD

Notch2 encodes one of four receptor genes (*Notch1–4*) involved in the highly conserved ligand–receptor notch-signaling pathway. Interaction of these cell surface receptors with extracellular notch ligand leads to protease cleavage of the C-terminal polypeptide known as the notch intracellular domain (NICD). NICD translocates to the nucleus and induces transcription of target genes through interactions with co-repressors and co-activators [31]. Apart from the canonical pathway used by *Notch1–4*, Notch2-ICD (N2ICD) is produced as an independent peptide through a cap-independent process involving an IRES element present within the coding region of the *Notch2* mRNA [32]. The IRES-based expression of N2ICD provides an additional mode of notch signaling under different regulation [32]. Additionally, this dual mechanism of expression might have implications in understanding the various conditions associated with notch-signaling dysregulation, especially given that widely different clinical conditions result from mutations in different domains of *Notch2* [33–37].

Her2: HER2 and HER2 CTF

Her2 is part of the ErbB/Her family of receptor tyrosine kinases implicated in cancers and neurodegenerative disease. *Her2* is a polycistronic gene that encodes a full-length 185-kDa HER2 protein and several independent C-terminal fragments (~70 kD HER2 CTF) [38]. Two methionines, one at position 611 and the other at 687, act as initiators of translation for CTFs. The existence of these internal initiation sites suggests the presence of an IRES within the coding region of HER2, although canonical IRES identification methods have not been conducted. The HER2 protein is a tyrosine kinase receptor known to act in secretory and endocytic pathways. The full-length protein also potentially localizes in the cell nucleoplasm. HER2 CTFs encompass the transmembrane and cytoplasmic domains of HER2 and may be involved in gene expression. Overactivation of *Her2* is implicated in hereditary breast cancer and cancer metastasis. Therapeutic administration of antibodies to surface HER2 is a mainstay of immunotherapy for breast cancer. However, HER2 CTFs expressed in the absence of HER2 surface receptors were shown to be unresponsive to therapeutic antibodies. These tumor cells were only sensitive to kinase inhibitors [38]. As a result, revelations in the polycistronic regulation of *Her2* cancer may allow for improved development and selection of cancer therapies.

Concluding Remarks

The organization and expression of specific proteins from mammalian polycistronic mRNAs appear to provide a similar layer of coordinated expression control to that used constitutively by invertebrates and protozoans, adding to other expression controls believed to be exclusive to mammals. Among the few examples of human polycistronic genes recognized, the additional cistrons appear to allow for: (i) co-expression of multiple subunits of a protein that is part of a complex; (ii) co-expression of similarly functioning proteins under slightly different spatial or temporal patterns; or (iii) co-production of novel, independently functioning but programmatically related proteins. Although little is understood at present about the cellular machinery involved in the regulation of these IRESs, or whether the upstream cistron product may have a role in the function of the IRES or expression of the second protein, these controls may have important implications in the understanding of health and disease (see Outstanding Questions). ITAFs have been shown to both promote and inhibit the translation of cellular IRESs, suggesting their dynamic role in regulating downstream cistron expression [6,39]. In addition, overexpression or depleted expression of ITAFs could cause direct dysregulation of the secondary protein that could result in disease. A possible therapeutic strategy could be to use genetic knockdown or drug intervention to restore the balance of ITAF expression [40]. Additionally, a disease state may result in alternative localization or alternative cell specificity of ITAFs and thereby change the regulation of secondary protein expression. This is a critical consideration to take into account when developing therapeutics against diseases involving genes with functional IRESs [41].

In cases where polycistronic activity is compromised through mutations in important regions within the IRES, therapeutics could target the expression of the second protein independent of the IRES (e.g., via adeno-associated virus injection of a cap-dependent form of the mRNA sequence of the secondary protein) or possibly genetic editing to restore IRES function. Additionally, if the secondary protein expression needs to be abrogated, the IRES may be targeted directly through RNAi (i.e., miRNA) or small-molecule inhibitors [29,40]. For example, selective inhibition of the function of an IRES may enable selective suppression of a disease protein, as shown in an animal model of SCA6, and could, in theory, be the case with persistent expression of HER2 intracellular protein in treatment-resistant breast tumors [29,38]. Additionally, second cistron expression of an antigen exclusively in cancer cells, such as in the case of *meloe*, presents an opportunity for immunotherapy [15,25]. Thus, if the IRES-specific mechanisms governing the expression of the additional cistrons are unique, they may be ideal drug targets. Furthermore, elucidating various IRES control mechanisms, such as eIFs, ITAFs, miRNAs, and other regulatory features, may provide insight into the mechanism of poly amino acid expression seen in diseases associated

Outstanding Questions

What additional polycistronic genes can be identified by candidate gene approach or non-biased screens that are regulated by IRES-mediated translation?

What are the molecular and cellular pathways that regulate these novel IRESs?

What off-target effects could arise if small molecule inhibitors or immunotherapy were used to target pathological secondary protein production in polycistronic genes?

Insights into mechanisms of alternative-translation control of these IRESs, such as initiation factors and ITAFs could reveal new sites for clinical treatment, particularly in diseases associated with RAN translation.

with **repeat associated non-ATG (RAN) translation** [42,43]. Finally, if the function of the upstream cistron-encoded protein has a role in the expression of the downstream protein, genetic aberrations of this gene could in theory have pleiotropic consequences. For example, because depolarization may affect the translocation of the *CACNA1C* calcium channel transcription factor, **CCAT**, to the nucleus, gain- or loss-of-function mutations of the calcium channel portion of this gene may affect the entry of CCAT into the nucleus and expression of the array of genes under its control [44]. Thus, further understanding of the control of polycistronic gene expression in vertebrate tissues should provide new insights into different human genotype–phenotype correlations as well as treatments of human disease.

Supplemental Information

Supplemental information associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tig.2016.11.007>.

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